

Intermediate Elective 2

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```
# general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)
library(RColorBrewer)

# reading in .xlsx files
library(readxl)


# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")
```

```
# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

#1. Explain

- I chose the most popular Ikea designers because I felt that it would be a good type of visualization to show the most abundance invertebrate species at NCOS.
- This visualization makes sense for this the invertebrate visualization because there are 15+ different designers, which is similar to how many different taxa there are.
- All Taxa will be shown in my visualization. There will be a different color for each taxon, and the amount of the color will show how abundant they show up in samples.

#2. Plan

```
knitr::include_graphics(here("data", "plan.jpeg"))
```



#3. Code

```
# creating new object from sites
ncos_sites <- sites |>
```

```
# filter to only include NCOS quarterly sampling sites
filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
water_quality_clean <- water_quality |>
```

```
# cleaning column names
clean_names() |>
```

```
# filtering join: only include sites in ncos_sites
# keep all the rows on the left that match up with rows on the right
semi_join(ncos_sites, by = "site") |>
```

```
# combine date and site together and remove those indiv. columns
unite("date_site", date, site, remove = TRUE) |>
```

```
# group by date_site
group_by(date_site) |>
```

```
# create new columns that show median of each variable at each date_site
# because there are multiple samples at the same site at different depths
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
```

```
# ungroup
ungroup() |>
```

```
# filter out the salinity outlier
filter(date_site != "2022-08-12_NVBR")
```

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
```

```
# converted taxon name values to snake case (lowercase)
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))
```

```

# creating new clean aquatic inverts object from aquatic inverts
aq_ins_clean <- aq_ins |>

# cleaning column names
clean_names() |>

# filtering join: only include quarterly NCOS surveys from ncos_sites
semi_join(ncos_sites, by = "site") |>

# renaming column to just date
rename(date = date_on_vial) |>

# select columns of interest
select(date, sample_type, site,

      # 9 most abundant taxon
      ostracod,
      copepod,
      hemiptera_corixidae_boatman,
      cladocera,
      nematode,
      diptera_ceratopogonidae,
      annelida_oligochaete,
      diptera_chironomid,
      annelida_polychaete) |>

# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250")) |>

pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)

```

```

summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

# create a new column called `date_site` to join with water quality
# don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
↪ aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>

# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
↪ = TRUE),
  site_full = fct_rev(site_full))

```

```

data <- aq_ins_clean |>

select(taxon, avg_per_lit) |>

group_by(taxon) |>

mutate(total_avg_abundance = sum(avg_per_lit)) |>

```

```

ungroup() |>

mutate(log_total_avg = log(total_avg_abundance + 1)) |>

select(taxon, log_total_avg) |>

unique() |>

# change taxon values to title case
mutate(taxon = str_replace_all(taxon, "_", " ") |> str_to_title())

data <- mutate_at(data, vars(taxon), as.factor)

glimpse(data)

```

```

Rows: 9
Columns: 2
$ taxon      <fct> Annelida Oligochaete, Annelida Polychaete, Cladocera, Co~
$ log_total_avg <dbl> 2.2823824, 2.1361369, 7.4709077, 7.9159808, 2.0541237, 2~

```

```

data <- data[order(-data$log_total_avg), ]
data$id <- seq(1, nrow(data))

data$id <- seq(1, nrow(data))

# Get the name and the y position of each label
label_data <- data
number_of_bar <- nrow(label_data)
angle <- 90 - 360 * (label_data$id-0.5) /number_of_bar
label_data$hjust <- ifelse(angle < -90, 1, 0)
label_data$angle <- ifelse(angle < -90, angle+180, angle)

pal <- brewer.pal(11, "Spectral")

# Plotting
p <- ggplot(data)+

  geom_bar(aes(x=as.factor(id), y=log_total_avg, fill=taxon),
    ↪ stat="identity") +

```

```

scale_fill_manual(values=pal)+

ylim(-40,90) +

theme_bw() +

theme_minimal() +

theme(
  axis.text = element_blank(),
  axis.title = element_blank(),
  panel.grid = element_blank(),
  plot.margin = unit(rep(-1,4,), "cm")) +

annotate(geom = "text",
  x=0,y= -30,
  hjust=.5, vjust=1,
  label="Taxon Abundance \n at NCOS",
  size=3, lineheight=.8,
  color="black") +

annotate(geom = "text",
  x=0,y=200,
  vjust=1,
  hjust=.5,
  label = "Data:  Kaggle & TidyTuesday | Graphic: Cristina Cametti -
  ↪ @CameCry",
  size=2.5,
  color="black") +

coord_polar(start = 0) +

geom_text(data=label_data, aes(x=id, y=log_total_avg + 1, label= taxon,
  ↪ hjust=hjust), color="black",
  fontface="bold",alpha=0.6, size=3.4, angle=label_data$angle,
  ↪ inherit.aes = FALSE )

```

p

